SIX UNIT SQUARES CANNOT COVER A LARGER SQUARE: AN ENGLISH CANDIDATE-PROOF MANUSCRIPT

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Abstract. Let $S(n)$ be the largest side length of a Euclidean square that can be covered by n closed unit squares, allowing arbitrary rigid motions and overlaps. Equivalently, Soifer's function $\Pi(2)$ is the least number of unit squares needed to cover *some* square of side $2+\varepsilon$ for *some* $\varepsilon > 0$. We present a computer-assisted *candidate* argument that $S(6) = 2$, or equivalently $\Pi(2) = 7$, addressing the conjecture stated by Dósa–Lángi–Tuza [4]. In particular, six unit squares cannot cover any square of side strictly greater than 2, while side 2 is attained by the obvious 2×2 tiling. The matching upper bound $\Pi(2) \leq 7$ is the classical fact that seven unit squares cover some square of side > 2; see Soifer and the covering surveys cited below.

The argument is a finite case analysis on ownership of the centre and of the four outer midpoints, followed by stepwise normalisation / reclassification and the elimination of joining modes. Algebraic exclusions are discharged by exact rational identities, positive Bernstein certificates on compact boxes, interval certificates, and SMT UNSAT queries; selected interfaces are sketched in a Lean 4 skeleton. Machine certificates support the narrative; Lean Boolean flags are status metadata, not geometric theorems. Independent peer review has not occurred; see the repository status notes for open gaps.

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Computer-assisted candidate manuscript; generative tools used in production. The releaser does not claim mastery of every step; independent peer review is pending. See the repository `DISCLOSURE.md`.

Guide for referees

This short paper is a computer-assisted *candidate* argument, not a peer-reviewed resolution. The detailed English appendix is not yet included. Read the core in this order: (i) Definitions and Proposition 3.3 (normalisation honesty); (ii) Proposition 4.1 and Appendix C (branch tables); (iii) Propositions in Sections 5–11, each stating *hypothesis* ⇒ *conclusion* and citing long-proof modules by ID; (iv) Appendix A–D (closure patches M1–M4); (v) repository `referee_pack/REFEREE_BRIEF.md` for timed reading paths. Boolean Lean flags are status metadata. Do *not* treat dual-solver UNSAT alone as a machine-independent proof.

Scope of this English edition

This draft establishes one logical route in English for the core argument. The complete English technical appendix is still in preparation and is not included in this release. Accordingly, this document is not a self-contained full proof: propositions that cite R-modules still require those detailed derivations. The historical source is pinned to commit `99408ee`. This is an editorial reconstruction of a candidate proof, not a new assertion of independent mathematical validation. Stable module identifiers locate the detailed source and arithmetic evidence. The technical translation requires mathematical editing; preserving formulae does not establish that a translation preserves every hypothesis or implication.

Appendix A gives the only active normalization protocol. The stronger one-shot global-clamping scheme is excluded from this reading route. Global notation is fixed before the geometry begins. Auxiliary parameters of a technical lemma are locally scoped: the same letter in two lemmas never licenses transferring a hypothesis.

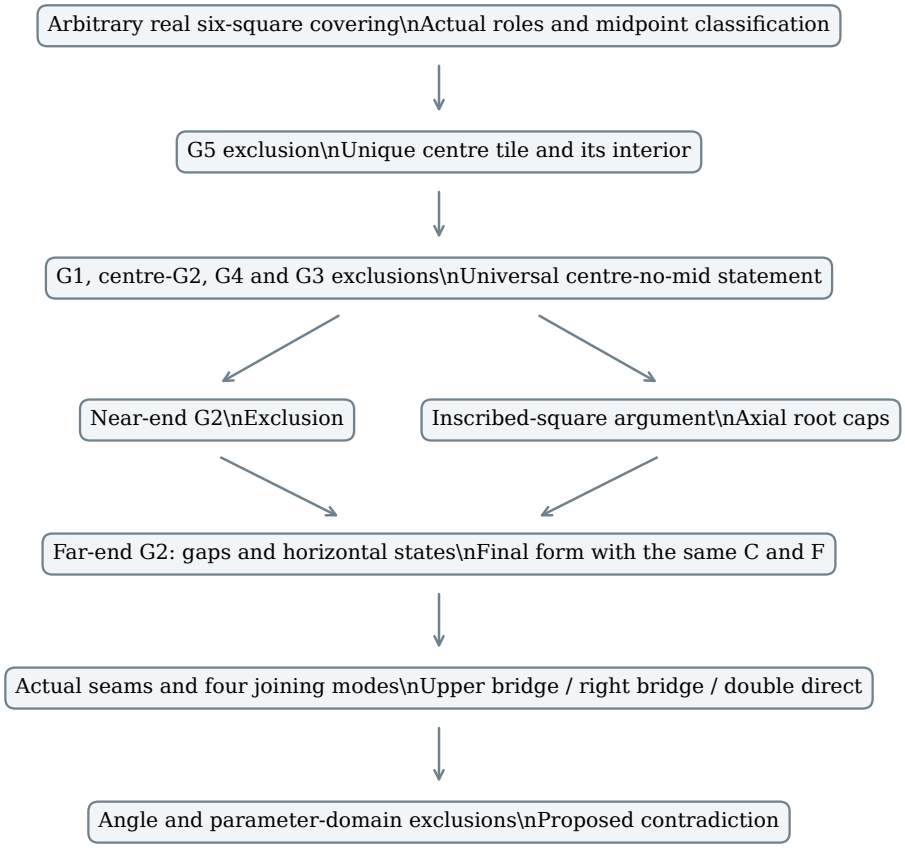


FIGURE 1. The proposed logical dependency order. An arrow is a geometric implication to check, not an arithmetic verification result.

1. INTRODUCTION

Covering a large square by equal smaller squares is a classical theme in discrete and metric geometry. Write

$$S(n) \;=\; \sup\bigl\{\, s>0 : \text{a closed square of side } s \text{ admits a covering by } n \text{ closed unit squares} \,\bigr\}, \tag{1}$$

where each unit square may be translated and rotated and overlaps are allowed. Soifer's function $\Pi(n)$ is the least number m of unit squares for which there *exists* some $\varepsilon>0$ such that a square of side $n+\varepsilon$ admits a covering by m unit squares (equivalently: there exists $\varepsilon_0>0$ such that every side in $(n, n+\varepsilon_0]$ admits such a covering); see [9, 8]. With this existential reading, and with the classical construction that seven unit squares cover some square of side strictly greater than 2 (hence $\Pi(2)\le 7$), one has

$$S(6)=2 \quad\Longleftrightarrow\quad \Pi(2)=7. \tag{2}$$

The lower bound $S(6)\ge 2$ is elementary: four axis-aligned unit squares tile $[0,2]^2$. The upper bound $S(6)\le 2$ is the content of the present paper; together with $\Pi(2)\le 7$ it would yield $\Pi(2)=7$.

Januszewski proved that covering a square of side $2+\varepsilon$ requires at least six unit squares, i.e. $\Pi(2)\ge 6$ [7]. Exact small values such as $S(5)=2$ appear in the literature of Friedman–Paterson and Januszewski; see also Friedman's covering surveys [6, 5]. In January 2026, Dósa, Lángi and Tuza revisited congruent square coverings, determined several small $S(n)$ and boundary-covering variants, and stated as Conjecture 1.1 that $S(6)=2$ (equivalently $\Pi(2)=7$) [4].

Theorem 1.1 (Proposed main theorem)**.** $S(6) = 2$. *Equivalently (given* $\Pi(2) \le 7$ *from known seven-square constructions),* $\Pi(2) = 7$*: six closed unit squares cannot cover any square of side strictly greater than* 2*, while a square of side* 2 *admits such a covering.*

The proposed proof route occupies Sections 2–11. Section 12 records the logical rôles of the computational certificates. Related work is collected in Section 13.

Remark 1.2 (Status)**.** Theorem 1.1 is offered as a *computer-assisted candidate.* This editorial draft and the selected certificate checks are not an independent verification of the full geometric chain. The main theorem has not been established in Lean. Boolean flags are not geometric theorems. The explicit review gates for this edition are listed in `REVIEW_OBLIGATIONS.md`.

2. Definitions and standing notation

Definition 2.1. For each integer $n \ge 1$, $S(n)$ is defined by (1). Soifer's $\Pi(n)$ is the least integer m for which there *exists* some $\varepsilon > 0$ such that a square of side $n+\varepsilon$ admits a covering by m unit squares. Equivalently, there exists $\varepsilon_0 > 0$ such that every square of side in $(n, n+\varepsilon_0]$ admits such a covering. (The universal quantifier "for every $\varepsilon > 0$" is incorrect: for fixed m, large ε violate the area bound $(n+\varepsilon)^2 \le m$.)

Throughout the upper-bound argument we work with the centred target

$$T \;=\; [-\mu, \mu]^2, \qquad \mu > 1, \tag{3}$$

and derive a contradiction from the assumption that six closed unit squares cover T. (The translate $[0, 2\mu]^2$ is equivalent for all length budgets.)

Write O for the centre of T, label the four corners $1, 2, 3, 4$ counter-clockwise from the lower left, and write BOTTOM, RIGHT, TOP, LEFT for the outer edges. An *outer mid* is the midpoint of an outer edge. The central horizontal and vertical target segments are denoted H_0 and V_0. A *true gap* on an outer edge is a nonempty open subinterval of that edge not covered by the two adjacent corner tiles; endpoints of gaps are never free hypotheses and must be recovered from openness, convexity and closedness.

"Corner tiles abut" on an outer edge means that the two adjacent corner tiles together cover the entire edge (overlaps allowed). Otherwise the uncovered open interval is a true gap.

Global notation.

Symbol	Meaning
$T_\mu = [-\mu, \mu]^2$	The original target, with $\mu > 1$.
$O;\ v_i;\ m_E$	Centre; corners numbered counterclockwise from lower left; midpoint of side E.
$V_i,\ C,\ F$	Four chosen corner hosts, a distinct centre host, and the remaining sixth tile. Hosts need not be exclusive.
W	The unanchored corner among the V_i in the current representative, never the non-corner tile F.
$H_0,\ V_0$	Central horizontal and vertical target segments.
κ	Number of distinct outer midpoints in $C \cup F$.
l, r, u, d	Actual left, right, upper and lower axial reaches of C.
L_C, R_C, U_C, D_C	$(-l, 0), (r, 0), (0, u), (0, -d)$.
a, x, z	Final-form half-angle parameters of V_1, V_2, V_4.
η, τ	Half-angle parameter of C and, when used, of W.
L_i, S_i, U_i	Anchored long outer leg, other outer leg, and central-axis inward contribution of V_i.
Σ	$S_1 + S_2$ in the final far-end configuration.
Q_R, Q_U	Positive-axis inward tips of V_2, V_4.

A named point and its distance from an axis are different objects. In particular U_i is a length, whereas U_C is a point. The point P_1 introduced later does not define a new covering with $\mu = 1$.

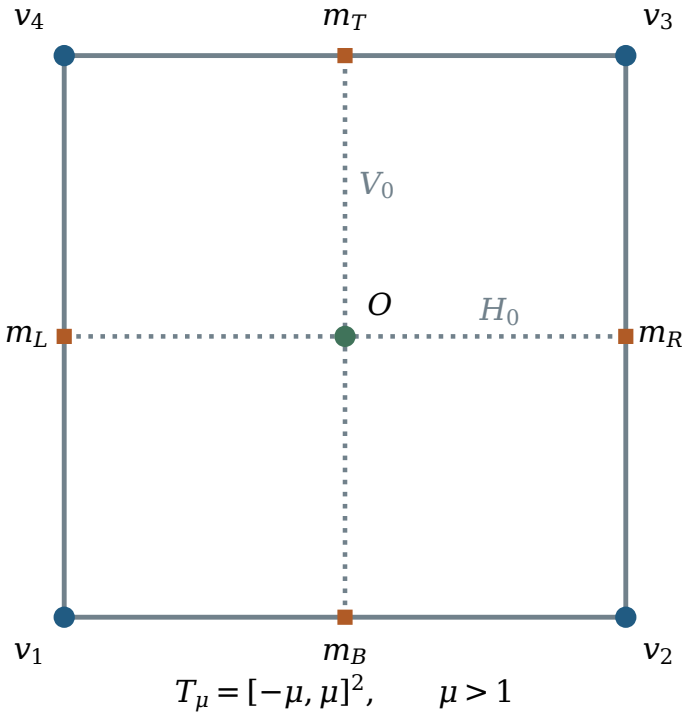


FIGURE 2. The fixed target and six-line grid. Corner indices determine the geometric roles; discovery-round indices are only source locators.

3. Roles, normalisation, and parallel budgets

Lemma 3.1 (Roles and midpoints)**.** *Four distinct tiles V_1, V_2, V_3, V_4 cover the four corners of T; a fifth tile C covers O; the sixth tile is denoted F. Each unit tile contains at most one outer mid. The role selection does not forbid F from containing a corner, nor a mid from lying in both a corner tile and a non-corner tile.*

Proof. Distinct target corners are at distance at least $2\mu > \sqrt{2}$, hence cannot lie in one unit tile. The distance from a corner to the centre is $\sqrt{2}\,\mu > \sqrt{2}$, so C is not a corner tile. Distinct outer mids are at least $\sqrt{2}\,\mu > \sqrt{2}$ apart. □

Lemma 3.2 (Dominated normalisation)**.** *Fix a target corner as origin with inward coordinates $X, Y \geq 0$. The unit tile containing that corner may be enlarged inside T to the form*

$$0 \leq cX + sY \leq 1, \qquad -\beta \leq -sX + cY \leq 1 - \beta, \quad c, s \geq 0,\ c^2 + s^2 = 1, \tag{4}$$

without moving any other tile and without deleting any previously covered point of T.

Proof. Among the four oriented unit normals choose one in the *closed* first quadrant (so $c, s \geq 0$; axis-aligned tiles have a normal on the boundary of the open first quadrant). Its original projection interval $[\alpha, \alpha + 1]$ contains 0, hence $\alpha \leq 0$. Replacing it by $[0, 1]$ does not remove any point of T with nonnegative projection. Keep the orthogonal unit interval unchanged. □

Proposition 3.3 (Dominated corner enlargement; stepwise reclassification)**.** *Starting from any covering of T, apply Lemma 3.2 to a corner tile V_i, leaving the other tiles fixed. The resulting covering still covers T and does not delete previously covered points of T. The same push applied a second time to that same tile is the identity on the already enlarged tile (idempotence of that single operator).*

This does not *imply that every later local normalisation or clamping step (for example, the $\beta \leq s/c$ clamp used for anchored corner tiles in the long verification manuscript) acts as the identity,*

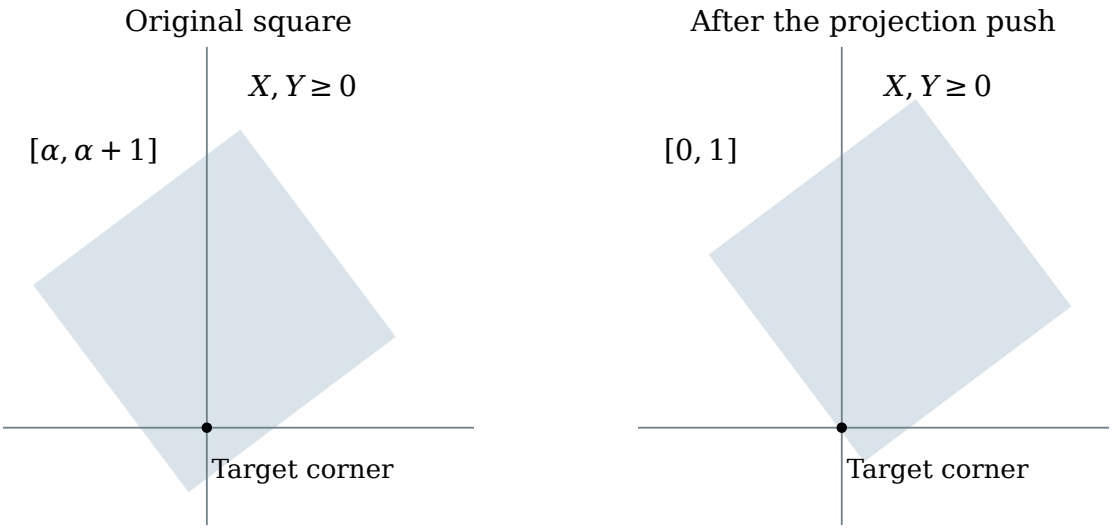


FIGURE 3. One projection push, with the orthogonal interval fixed. Containment holds inside the inward target quadrant; it is not asserted between the two whole squares in the plane.

nor that classification flags computed before an enlargement remain valid afterwards. After each enlargement that may create duplicate hosts or fill gaps, the covering must be reclassified *on the new configuration; see the round 65 audit notes on β-clamping and stepwise reclassification in the ancillary package (`artifacts/still_S/S6_round65_full_audit.md`, ğ4).*

Lemma 3.4 (Two nonempty parallel chords)**.** *If a unit tile meets two parallel lines at distance $d > 1$, then $d \leq \sqrt{2}$ and the sum of the two chord lengths is at most*

$$B(d) = \begin{cases} d - \sqrt{d^2 - 1}, & 1 < d \leq 3/(2\sqrt{2}), \\ 2(\sqrt{2} - d), & 3/(2\sqrt{2}) \leq d \leq \sqrt{2}. \end{cases} \tag{5}$$

In particular $B(d) < 1$. Tangency counts as nonempty; an empty chord may not be substituted into the bound.

Proof sketch. With direction parameters $c \geq s > 0$ the transverse width is $w = c + s$. The two vertical lines lie at t and $t + d$ with $0 \leq t \leq w - d$. Since $d > 1 \geq c$, both chords fall in the triangular end-caps and the total length equals $(w-d)/(cs) = 2(w-d)/(w^2-1)$. Maximising over $d \leq w \leq \sqrt{2}$ yields the two branches of (5). This geometric derivation, including the nonempty-chord premise, must be checked independently of the scalar maximisation; no completed Lean verification of this chord lemma is asserted here. □

Lemma 3.5 (Endpoints of a singly filled gap)**.** *If a nonempty open segment (A, B) lies in one fixed closed tile K, then both A and B belong to K.*

Proof. Take sequences in the open segment converging to its two endpoints and use closedness of K. If only $K_1 \cup K_2$ is known to contain the gap, the argument does not put both endpoints in either specified tile. □

Further elementary budgets (grid contribution at most 2 per corner tile; $L + S + U < 2$ for a normalised corner tile; triangular area at most $1/2$) are collected as Lemmas R2–R3 in the long verification manuscript and used below without restating the proofs.

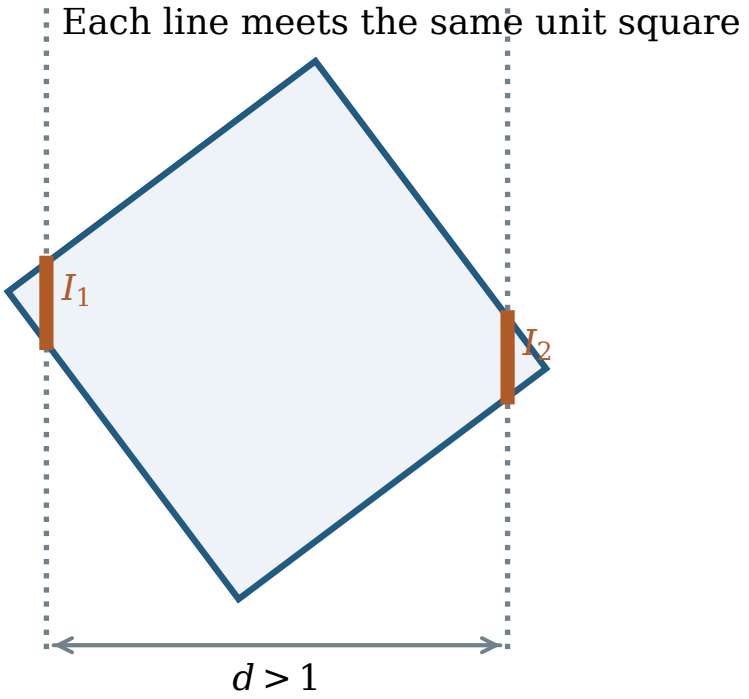


FIGURE 4. Two nonempty parallel chords of the same unit square. A tangent point is nonempty even though its chord length is zero.

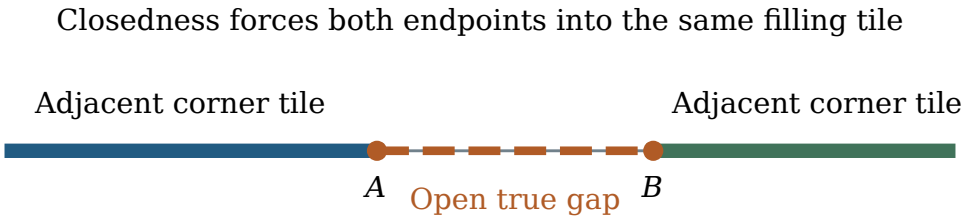


FIGURE 5. The same filling tile must first be identified for the entire open gap. Only then does closedness give both endpoints.

4. TOP-LEVEL CLASSIFICATION

If F also contains O, call the configuration class G5. Otherwise set

$$\kappa = \#\{\text{outer mids belonging to } C \cup F\} \in \{0, 1, 2\}. \tag{6}$$

Then $\kappa = 0$ is G1, $\kappa = 1$ is G2, and $\kappa = 2$ with adjacent (resp. opposite) mids is G4 (resp. G3). Each mid not yet supplied by $C \cup F$ chooses an adjacent corner host injectively. Reflection reduces host representatives.

Proposition 4.1 (Exhaustive branch table)**.** *Every covering of T falls into exactly one of the classes G5, G1, centre-owning G2, G4, G3, or residual G2 ($\kappa = 1$ with the unique mid owned by F on* BOTTOM*). The residual G2 splits into a near-end host representative* $(\text{RIGHT}, \text{TOP}, \text{LEFT}) = (2, 3, 4)$ *and a far-end representative* $(2, 4, 1)$.

Geometric exhaustiveness and host representatives. Every tile contains at most one outer midpoint, so the union $C \cup F$ contains at most two distinct such points. If F contains O, assign G5 before inspecting this count. Otherwise $\kappa = 0, 1, 2$ gives the asserted alternatives. When $\kappa = 1$, membership in C takes precedence, including shared membership in $C \cap F$; only exclusive membership in F enters residual G2. For $\kappa = 2$, the two midpoints are either adjacent or opposite, and each belongs to a different non-corner tile.

A midpoint not contained in $C \cup F$ must have a corner host among the same four chosen corner tiles. A target corner not adjacent to its side is at distance $\sqrt{5}\,\mu > \sqrt{2}$ from that midpoint, and so cannot be its host. Distinct midpoints cannot choose the same tile, by the unit-square diameter bound. Thus choosing one available adjacent host for each unsupplied midpoint automatically gives an injective host map; multiple available hosts are not discarded.

For residual G2 place the F midpoint on BOTTOM by a target symmetry. The left, top and right midpoint hosts belong respectively to $\{1, 4\}, \{3, 4\}, \{2, 3\}$. Removing repeated host indices leaves precisely:

RIGHT host	TOP host	LEFT host	Representative
2	3	4	near, $W = V_1$
3	4	1	reflection of near
2	4	1	far, $W = V_3$
2	3	1	reflection of far

Reflection in the vertical central axis fixes the bottom midpoint and pairs the indicated rows. It need not preserve a chosen corner index; indices are relabelled by the reflected target geometry.

For adjacent owned midpoints, put those of C, F on LEFT and BOTTOM. The two unsupplied hosts have $(\text{TOP}, \text{RIGHT}) \in \{(4, 3), (4, 2), (3, 2)\}$. For opposite owned midpoints, put those of C, F on LEFT and RIGHT. The four choices of bottom and top hosts reduce under horizontal reflection to the three representatives $(\text{BOTTOM}, \text{TOP}) = (2, 3), (1, 4), (1, 3)$. These are host choices in actual coverings, not Boolean assumptions about a newly invented layout. □

The global elimination order is: G5 for all three values of κ; then G1, centre-owning G2, G4 and G3; then near-end residual G2; then reduction of the far-end representative to the final standard form; then elimination of all joining modes of that form.

Remark 4.2 (Branch tables)**.** Explicit tables (M3.0–M3.3) and the Boolean near-end checker appear in Appendix C and `artifacts/closure/`. The short-paper propositions below state hypothesis ⇒ conclusion and cite long-proof module IDs; they are not self-contained geometric proofs of every leaf.

5. Elimination of G5, G1, centre-G2, G3 and G4

Proposition 5.1 (G5 impossible)**.** Hypothesis. *F also contains O (class G5), with $\kappa = \#\{\text{mid} \in C \cup F\} \in \{0, 1, 2\}$.* Conclusion. *No such covering exists; consequently in any covering of T exactly one tile contains O, and O is interior to that tile.*

Proof outline. Long-proof modules: R2 ($\kappa = 2$), R3 ($\kappa = 0$), R4–R8 ($\kappa = 1$). For G5, modules R2–R3 eliminate $\kappa = 2$ and $\kappa = 0$. For $\kappa = 1$, modules R4–R7 reduce to a triple-abutment configuration with an interior second centre tile; module R8 then proves the two-corner inequality

$$L_1 + L_2 + \min(t_1, t_2) + \tfrac{4}{5} \max(t_1, t_2) \le \tfrac{14}{5} \tag{7}$$

by a Bernstein certificate on forty-five rational rectangles after mapping the full two-dimensional domain into the unit square, and compares it with the weighted three-ray bound on the interior tile to reach a contradiction in both orderings of (t_1, t_2). Thus G5 is empty. The remaining five tiles stay at positive distance from O, so a small neighbourhood of O is covered by C alone and O is interior to C. □

Hyp	Modules	Concl
G5, $\kappa = 2$	R2	grid contribution of $C \cup F < 4$
G5, $\kappa = 0$	R3	residual-gap split empty
G5, $\kappa = 1$	R4–R7	reduce to interior second centre / triple abutment
same leaf	R8 + Bernstein cert	inequality (7) vs three-ray bounds

Detailed derivations. See the source modules cited above. Their detailed English appendix is still in preparation.

Lemma 5.2 (The closed central box)**.** *Once G5 is excluded for every target of side greater than two, the chosen central tile contains* $[-(\mu-1), \mu-1]^2$.

Proof. Let $K = (-\mu+1, \mu-1)^2$. Every $p \in K$ is the centre of some axis-aligned square with half-side strictly greater than one lying in T_μ. The same six tiles cover this inscribed square. Universal G5 exclusion says exactly one tile contains p. Consequently the six intersections of the tiles with K form a disjoint finite relatively closed partition of K. Each member is also relatively open, since its complement is a finite union of relatively closed members. Connectedness of K permits only one nonempty member. It is the member containing O, namely $C \cap K$. Thus $K \subset C$ and closedness gives $\overline{K} \subset C$.

The use of an inscribed square permits its corner hosts to be chosen afresh. This argument does not assume the later centre-no-mid theorem or root cap. □

Proposition 5.3 (G1 impossible)**.** Hypothesis. *Unique centre tile C (after Proposition 5.1) and $\kappa = 0$ (no outer mid in $C \cup F$).* Conclusion. *No such covering exists; hence $C \cup F$ contains at least one outer mid.*

Proof outline. Module R9 eliminates triple abutment by a quarter-boundary budget and reduces to opposite or adjacent residual gaps. For opposite gaps, the corrigendum R10C first forces the horizontal chord of F to be strictly positive (via $g_1 + h_C > 1$ against $B(\mu) < 1$), then combines parallel-chord budgets of C and F with the corner contribution bound < 8. Adjacent gaps are eliminated in R10 §§3–6 by real endpoints and direction constraints. □

Hyp	Modules	Concl
$\kappa = 0$, triple abutment	R9	impossible by quarter budget
$\kappa = 0$, opposite gaps	R10C + R10	$h_F > 0$ then parallel budgets conflict
$\kappa = 0$, adjacent gaps	R10 §§3–6	real endpoints / directions empty

Proposition 5.4 (Centre cannot own the unique mid)**.** Hypothesis. $\kappa = 1$ *and the unique outer mid in $C \cup F$ belongs to C (centre-G2), including the shared-ownership subcase $C \cap F \ni$* mid. Conclusion. *No such covering exists; the unique mid must belong to F alone.*

Proof outline. Axial propagation (R11) yields necessary outer contacts. Module R12 treats the two host representatives when only C owns the mid, covering all true-gap and triple-abutment patterns on LEFT, TOP and RIGHT. Shared ownership $C \cap F$ of the same mid is eliminated separately in R12 §10 by vertical redundancy and a true-gap triangle constraint. □

Module IDs: R11 (axial contacts); R12 (host representatives); R12 §10 (shared mid).

Proposition 5.5 (G4 impossible)**.** Hypothesis. $\kappa = 2$ *with C, F owning distinct* adjacent *outer mids (class G4), normalised so $C \supset$* LEFT mid *and $F \supset$* BOTTOM mid. Conclusion. *No such covering exists.*

Proof outline. The three injective host representatives $(\mathrm{TOP}, \mathrm{RIGHT}) \in \{(4,3),(4,2),(3,2)\}$ are eliminated in modules R20–R23 by horizontal-sign analysis, forced suspension segments, and projection budgets on C. Shared left-mid ownership by the free corner tile is ruled out before the last representative (R20A). □

Host (TOP, RIGHT)	Modules	Notes
$(4,3)$	R20, R21	all F-horizontal signs
$(4,2)$	R20, R21A, R22	suspension $J > x/3$
$(3,2)$	R20A, R22A, R23	shared left mid first

Proposition 5.6 (G3 impossible)**.** Hypothesis. $\kappa = 2$ *with* C, F *owning distinct* opposite *outer mids (class G3), normalised so* $C \supset$ LEFT mid *and* $F \supset$ RIGHT mid. Conclusion. *No such covering exists.*

Proof outline. Three host representatives (after reflection) are eliminated in R24–R25 by abutment/gap analysis on TOP and BOTTOM, empty/nonempty vertical chords of F, and a local exit lemma for nonempty cases (R24A/R25). □

Module IDs: R24, R24A, R25 (three host representatives).

Corollary 5.7 (Universal centre forbids outer mids)**.** *In any covering of* T *the unique centre tile contains no outer mid, and the remaining non-corner tile contains some outer mid.*

Proof. Combine Propositions 5.1–5.6. □

Lemma 5.8 (Inscribed-square axial cap)**.** *After the universal centre-no-mid conclusion, the actual axial reaches satisfy*

$$\mu - 1 \le l, r, u, d \le 2 - \mu < 1.$$

Proof. The lower bounds follow from Lemma 5.2. The centre-no-mid conclusion gives $r < \mu$. Suppose $r > 2 - \mu$ and set

$$\nu = \frac{\mu + r}{2} > 1, \qquad T' = [-\mu, r] \times [-\nu, \nu], \qquad p = \left(\frac{r - \mu}{2}, 0\right).$$

This is a square contained in T_μ, and $|p_x| = (\mu - r)/2 < \mu - 1$ gives $p \in C$. The same C contains its right midpoint $(r, 0)$. Apply the already universal centre-no-mid conclusion to T', with new corner hosts if necessary. This is impossible. Symmetry proves the other three upper bounds. No assumption that the original F remains a non-corner tile for T' is used. □

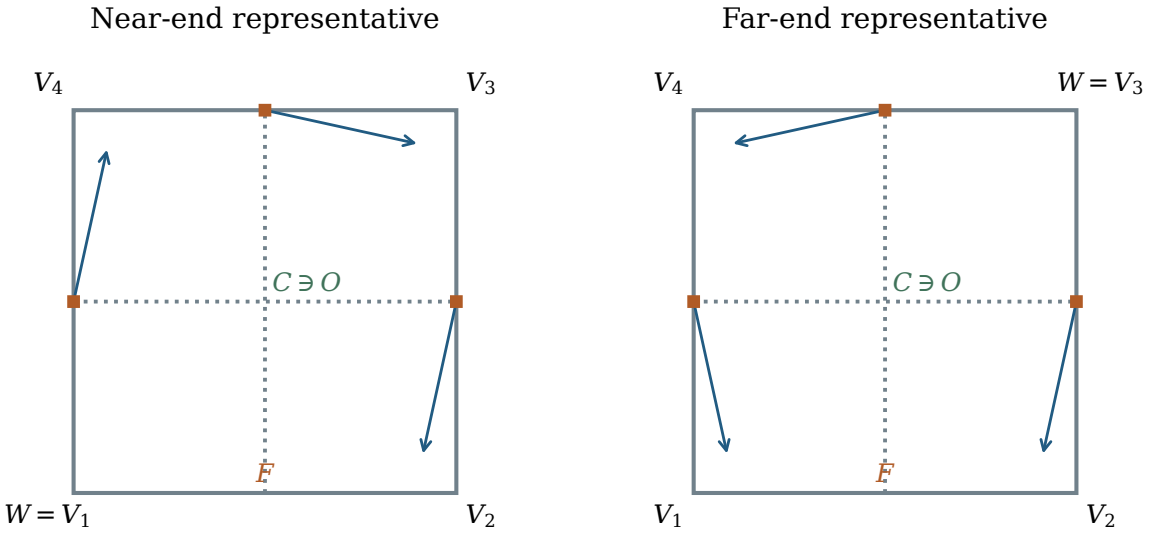


FIGURE 6. Residual host representatives after putting the midpoint in F on the bottom edge. Arrows indicate chosen membership, not exclusive ownership. The free corner is V_1 on the left and V_3 on the right.

6. Near-end residual G2

After Corollary 5.7 the only surviving class is residual G2 with the unique mid owned by F on BOTTOM.

Proposition 6.1 (Near-end impossible)**.** Hypothesis. *Residual G2 with F owning the bottom mid, C owning no outer mid, and host injection* $(\mathrm{RIGHT}, \mathrm{TOP}, \mathrm{LEFT}) = (2, 3, 4)$ *(free corner $W = V_1$).* Conclusion. *No such covering of T exists.*

Proof outline. See long-proof modules R13–R19 and R26–R28, summarised by the Boolean branch table in Appendix C (Table M3.1). Module R18 shows that either both RIGHT and TOP abut, or exactly one of them carries a true gap owned by C. In the double-abutment case: R13 eliminates nonempty horizontal chords of F; R14–R15 force left abutment when that chord is empty and produce an interior point Q_0 of C; R15S/R16P supply the strict angular separation and height hypotheses; R16 then covers the necessary continuous domain by 1173 rational boxes with interval certificates excluding every leaf. Remaining left-gap / nonempty-F patterns are eliminated in R17–R18. Top-gap patterns fall to R19 and R26; right-gap patterns fall to R27–R28 (both rotation signs). A normalisation that fills a gap returns to an already eliminated abutment leaf (Appendix A). □

Branch (Table M3.1)	Modules	Concl
RIGHT+TOP abut; F horiz. nonempty	R13	empty
same; F horiz. empty	R14–R16 (+R15S/R16P)	1173 interval boxes
LEFT gap / nonempty F	R17–R18	empty
TOP true gap	R19, R26	both rotation signs
RIGHT true gap	R27–R28	both rotation signs

Detailed derivations. See the source modules cited above. Their detailed English appendix is still in preparation.

7. Far-end reduction to standard form

Henceforth fix the far-end representative: V_1 owns the left mid, V_2 the right mid, V_4 the top mid, and the free corner tile is $W = V_3$. The centre tile C owns no outer mid; F owns the bottom mid and not the centre.

Proposition 7.1 (Far-end standard form)**.** Hypothesis. *Residual G2 with F owning the bottom mid, C owning no outer mid, and far-end hosts $V_1 \supset \mathrm{LEFT\,mid}$, $V_2 \supset \mathrm{RIGHT\,mid}$, $V_4 \supset \mathrm{TOP\,mid}$, free corner $W = V_3$.* Conclusion. *The covering satisfies:* LEFT, RIGHT *and* TOP *abut via corner tiles; $F \subset \{y < 0\}$; C avoids* BOTTOM*; and one copy of F contains both closed endpoints of the bottom gap together with the lower axial tip of C (the* final standard form*).*

Proof outline. See Table M3.2 in Appendix C and frozen module R29N in Appendix B.

Step 1. Modules R27 and R29 eliminate standardised top true gaps for both rotation signs of C. Thus TOP abuts. Any lateral true gap lies too far from the bottom mid for F to reach, so lateral gaps are owned entirely by C. Since C cannot meet both LEFT and RIGHT at distance $2\mu > \sqrt{2}$, the only remaining cases are left gap, right gap, or triple abutment (Proposition R29N.1).

Step 2. If $F \cap H_0 \neq \varnothing$, Proposition R29N.2 forces C to avoid BOTTOM; R31 eliminates triple abutment and R32–R36 eliminate the lateral-gap cases. Hence $F \cap H_0 = \varnothing$. If F contained a point with $y \geq 0$, convexity with $(0, -\mu) \in F$ would give $(q, 0) \in F$. Its diameter gives $q^2 + \mu^2 \leq 2$, and $\mu > 1$ implies $|q| < \mu$. This point therefore belongs to H_0, a contradiction. Thus the whole original tile satisfies $F \subset \{y < 0\}$.

Step 3. Corollary 5.7 applied to an inscribed target yields axial-root caps $\leq 2 - \mu$ (module R37) without circularity. Modules R37–R39 eliminate all right and left true-gap patterns (both rotation signs, direct and bridged contacts).

Step 4. In the residual triple-abutment case, R39 gives the upper axial root $u > 38/79$, whence C avoids BOTTOM. The three anchored corner tiles can contribute only their matching central half-axes, so the entire bottom gap and the lower tip of C are forced into the same tile F. □

Step	Modules	Concl
1	R27, R29, R29N.1	TOP abuts; only LEFT/RIGHT gap or triple abutment
2	R29N.2, R31–R36	$F \cap H_0 = \varnothing$ thence $F \subset \{y < 0\}$
3	R37–R39	no LEFT/RIGHT true-gap survivors
4	R39 ($u > 38/79$)	final standard form (bottom gap+C tip in F)

Detailed derivations. See the source modules cited above. Their detailed English appendix is still in preparation.

8. Standard-form notation and four joining modes

Let $a, x, z \in (0,1)$ be the half-angle parameters of V_1, V_2, V_4, and let $\eta \in [0, \sqrt{2}-1]$ be the half-angle of the smaller axial angle of C. Set

$$b(t) = \frac{1+t^2}{1-t^2}, \quad \lambda(t) = \frac{2t}{1-t^2}, \quad g(t) = \frac{1-t}{1+t}, \quad q(t) = \frac{t}{2-t}, \tag{8}$$

and write $\delta(t) = b(t) - 1$, $\Delta_z = \delta(z)$. The four actual axial tips of C are $L_C = (-l, 0)$, $R_C = (r, 0)$, $U_C = (0, u)$ and $D_C = (0, -d)$. Writing S_1, S_2 for the actual short legs of the bottom corner tiles and $\Sigma = S_1 + S_2$, Proposition 7.1 yields

$$\Sigma > 2 - \sqrt{2} > \tfrac{7}{12}, \qquad \Sigma > (1-d)^2, \qquad \Sigma > 2 - \frac{1}{1-d}. \tag{9}$$

The three lower bounds use one and the same original tile F. Its actual bottom chord has length $b_F \geq 2\mu - \Sigma$, and it contains the lower axial tip D_C. Its diameter gives $b_F \leq \sqrt{2}$, hence $\Sigma \geq 2\mu - \sqrt{2} > 2 - \sqrt{2}$. The three-axis unit-square inequality applied to this chord and tip gives $b_F + (\mu - d)^2 \leq 2$, so $\Sigma \geq 2\mu - 2 + (\mu - d)^2 > (1-d)^2$. For the last bound, the triangle with the two bottom-gap endpoints and D_C has area at most $1/2$:

$$(2\mu - \Sigma)(\mu - d) \leq 1.$$

Since $d < 1 < \mu$, it follows that $\Sigma \geq 2\mu - 1/(\mu - d) > 2 - 1/(1-d)$. The chord endpoints and the axial tip cannot be supplied by different placements of F.

Left abutment supplies the constraints

$$0 < z < a < 1, \quad a + 2z < 1, \quad z < \tfrac{4}{15}, \quad 0 \leq S_1 < \mathcal{F}(a, z) := b(a) - \lambda(a)\big(2 - g(z)\big). \tag{10}$$

Lemma 8.1 (Normalised seam point in C)**.** *There is a point*

$$P_1 = (-1 + X, e) \in C, \qquad X = \frac{b(a) + S_4 - 2}{\lambda(a) - \lambda(z)}, \quad e = 1 - S_4 - \lambda(z)X, \tag{11}$$

obtained from the real-scale seam $P_\mu \in C$ by a convexity argument inside the same tile C (modules R40 and R42). The construction does not rescale the covering to $\mu = 1$ and does not move C.

The unit-scale seam remains in the original tile. Put $k_a = \lambda(a)$, $k_z = \lambda(z)$ and $S = S_4$. The real-scale seam construction supplies

$$X_\mu = \frac{b(a) + S - 2\mu}{k_a - k_z}, \qquad e_\mu = \mu - S - k_z X_\mu, \qquad P_\mu = (-\mu + X_\mu, e_\mu) \in C.$$

Its membership is the actual geometric exit argument in technical lemma R40, not a claim that the intersection of two extended supporting lines automatically belongs to C. The negative horizontal corner endpoint also belongs to C:

$$Q_L = (-\ell_\mu, 0), \qquad \ell_\mu = \mu - a + \frac{\mu - 1}{k_a}.$$

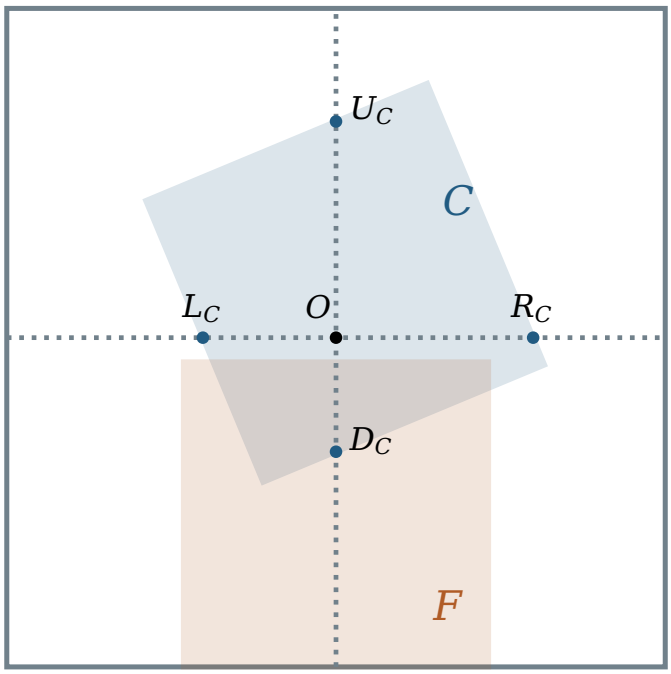


FIGURE 7. Actual axial tips of one illustrative rotated unit tile, and an illustrative unit tile below H_0. Only two tiles are shown; no six-tile covering is asserted. The argument requires the same F to contain the bottom-gap endpoints and D_C.

The support identities give $0 < X_1 < a$ and $e_1 > k_z(1 - z - a) > 0$, together with

$$e_\mu - e_1 = (\mu - 1)\left(1 + \frac{2k_z}{k_a - k_z}\right) > 0.$$

Since $P_\mu \in V_1$ and V_1 contains the lower-left target corner, $e_\mu + \mu \le \sqrt{2}$. The already established $u > 38/79$ yields

$$0 < e_1 < \sqrt{2} - 1 < 5/12 < 38/79 < u,$$

so $(0, e_1) \in C$. The first upper support of V_1 gives the exact identities

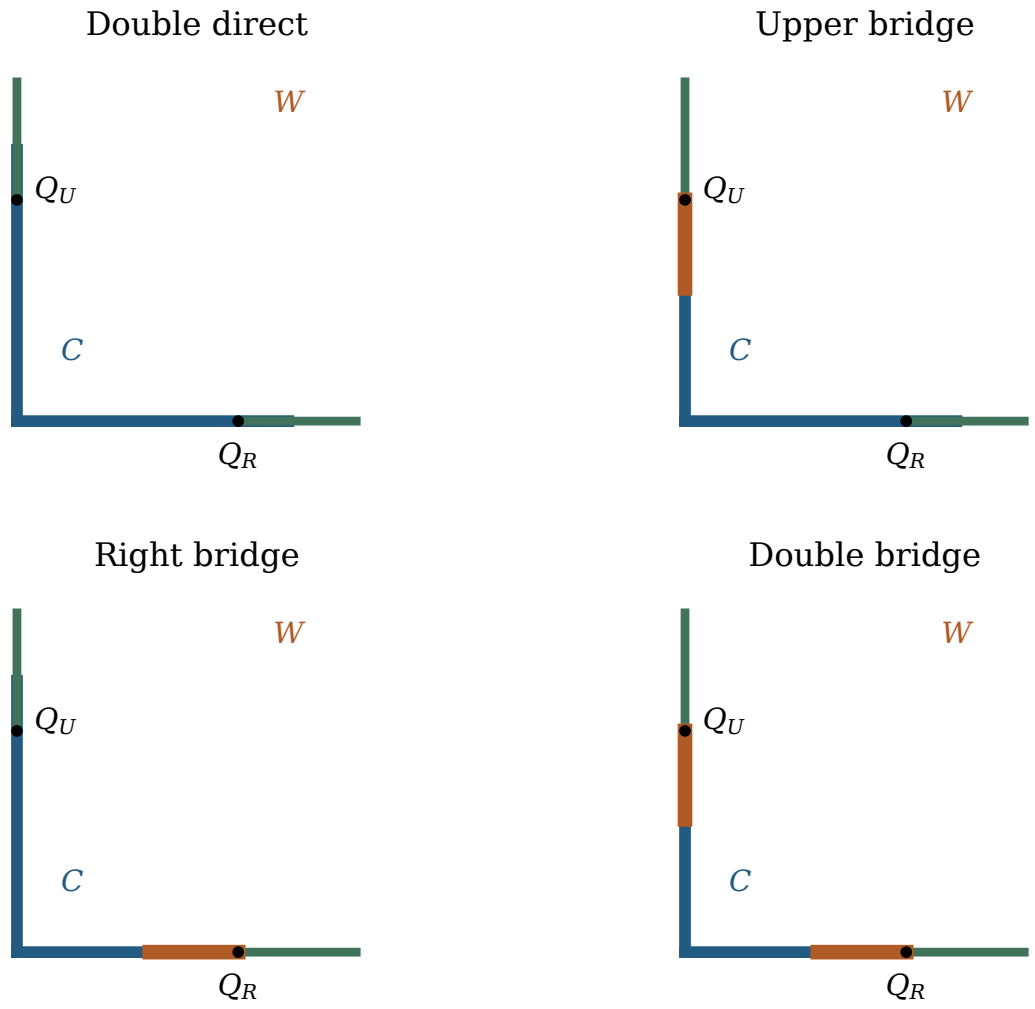
$$\mu - X_\mu = \ell_\mu + \frac{e_\mu}{k_a}, \qquad 1 - X_1 = (1 - a) + \frac{e_1}{k_a}.$$

The segment $Q_L P_\mu \subset C$ contains, at height e_1,

$$Z = (-\ell_\mu - e_1/k_a, e_1).$$

Since $\ell_\mu - (1 - a) = (\mu - 1)(1 + 1/k_a) > 0$, the point $P_1 = (-1 + X_1, e_1)$ lies between Z and $(0, e_1)$. Convexity proves that P_1 belongs to the same original C. No tile, target, or family has been rescaled to $\mu = 1$. □

Let Q_R and Q_U be the actual inward tips of V_2 and V_4 on the positive axes. Whether C contains these tips yields four exhaustive joining modes: double-direct, upper bridge by W, right bridge by W, and double bridge. If C misses a tip, the corresponding open axial gap can be filled only by the same free corner tile W, which is then forced by closedness to contain both the tip of C and the corner tip.



Blue: central axial interval. Green: anchored corner interval. Orange: a gap forced into W.

FIGURE 8. The four membership alternatives for Q_R, Q_U. A missing direct membership forces the open axial gap, and then its endpoints, into the same original W. These are interval incidence diagrams, not purported six-tile coverings.

9. Elimination of upper bridges and negative rotations

Proposition 9.1. *In the standard form of Proposition 7.1, the free corner tile W cannot contain Q_U.*

Proof outline. Modules R44–R45 show that an upper bridge forces $z \le \tau < 2/5$ for the parameter τ of W and empties $W \cap H_0$, whence C contains Q_R directly and double bridging is impossible. Modules R46 and R48 produce a real exit point $T_R \in C \cap W \cap V_2$. Small angles $\eta \le 1/24$ fall to R52. Positive rotations are eliminated in R53–R55 via the seam-derived bound $S_1 < q(Q)$ and the two normals of C at T_R. Negative rotations are reduced in R56–R58 to $0 < \tau < 1/6$ and then excluded in R59 by the area-forced negativity

$$G = H_F - q(x+\delta) - g(x) + \lambda(x)(B_t - 1) < 0, \quad H_F = 2 - \frac{1}{1-\delta-\tau}, \tag{12}$$

using endpoint polynomial certificates on small/medium x and a left-root projection contradiction on large x. □

Consequently only double-direct and right-bridge modes remain, both with $C \ni Q_U$ and hence $u > 1 - z$. In the right-bridge case the horizontal chord length of W is strictly less than Δ_z, which after establishing that the relevant parameters are < 1 yields

$$S_2 < q(r + \Delta_z); \tag{13}$$

double-direct admits the stronger $S_2 < q(r)$.

Proposition 9.2. *All negative rotations in the two residual joining modes are impossible. Every surviving positive rotation satisfies* $1/10 < z < 1/5$.

Proof outline. Small-angle double-direct and right-bridge cases are eliminated in R43 and R49–R51 respectively, both forcing $\eta > 1/8$. Module R60 partitions the negative-rotation domain into four regions and contradicts (9) in each, paying the loss (13). Modules R61–R62 eliminate $z \geq 1/5$ and $z \leq 1/10$ for positive rotations by asymmetric triangles on the same F and concavity of the budget form, leaving the open strip $1/10 < z < 1/5$. □

10. Right bridges are impossible

Proposition 10.1. *In the residual domain of Proposition 9.2, W does not contain the right axial tip R_C of C. Consequently the entire right-bridge class is empty.*

Proof outline. Module R63 first obtains $r < 17/20$ from axial chords and the R/U_C projection. Assuming $W \ni R_C$, the tile meets both TOP and H_0, forcing a top leg $D > 1 - \Delta_z > 11/12$ and a half-angle

$$t < \tfrac{1}{8}, \qquad t < \tfrac{9}{7}\Delta_z. \tag{14}$$

Comparing real projections in both orientations of W yields an abscissa ξ of the contact on H_0 satisfying

$$\xi < \Delta_z + t \quad \text{or} \quad \xi > 1 - t > \tfrac{7}{8}. \tag{15}$$

The second alternative contradicts $r < 17/20$ if the contact is R_C. The first, combined with (13)–(14) and left long-leg loss, produces

$$\Sigma < \Phi\Big(\frac{2z}{1+z}\Big) + q\Big(\frac{23}{7}\Delta_z\Big), \quad \Phi(w) = (1+w)\big(1 - \sqrt{w(2+w)}\big), \tag{16}$$

which is decreasing on $1/10 \leq z \leq 1/5$ and strictly less than $7/12$ at the left endpoint, contradicting (9). □

11. Double-direct positive rotations

Proposition 11.1. *No positive-rotation double-direct covering survives in the residual strip of Proposition 9.2.*

Proof outline. From (10) and $1/10 < z < 1/5$ one obtains $2/7 < a < 3/5$. Double-direct gives $S_2 < q(r)$. At the formal endpoint $S_4 = g(z)$ define

$$X_G = \frac{(1-z)\big[a^2(1+2z) - z\big]}{(a-z)(1+az)}, \qquad e_G = \frac{2az(1-a-z)}{(a-z)(1+az)}. \tag{17}$$

After verifying $a < g(\eta)$ one has the projection bound

$$r < R_P(\eta) := \delta(\eta) + X_G + \lambda(\eta) e_G, \tag{18}$$

while the pair (U_C, R_C) yields $r < R_T(\eta) := \eta + z/\lambda(\eta)$ for $\eta > 0$.

On $0 \leq \eta \leq 1/4$ one has $a < 3/5 \leq g(\eta)$; monotonicity of (18) reduces the right-root upper bound to $\eta = 1/4$, after which $\mathcal{F} + q\big(R_P(1/4)\big) < 7/12$ on the full (a, z)-rectangle. On $1/4 \leq \eta \leq 1/3$, concavity of the real P_1/U_C relation forces $a > 2/5$, and four endpoint values control $\Sigma < 7/12$. On $1/3 \leq \eta \leq \sqrt{2} - 1$, a rational outer cap $29/70$ splits according as $a \geq g(\eta)$ or not; both subcases again force $\Sigma < 7/12$.

The fourteen deduplicated rational targets are certified by twenty-eight numerator and denominator polynomials whose one hundred fourteen strictly positive rational Bernstein coefficients, together with reconstruction identities, cover the whole continuum $[0, \sqrt{2} - 1]$ (module R64). Every angular segment contradicts (9). □

Proof of Theorem 1.1. Assume six unit squares cover T with $\mu > 1$. Proposition 4.1 places the covering in one of the top-level classes. Propositions 5.1–5.6 eliminate G5, G1, centre-owning G2, G3 and G4. Proposition 6.1 eliminates the near-end residual G2. Proposition 7.1 reduces the far-end residual G2 to the final standard form. Propositions 9.1–11.1 eliminate every joining mode, rotation sign and residual parameter domain of that form. Hence (subject to the completeness of the cited modules and certificates) $\mu > 1$ is impossible under the candidate narrative. Combined with the elementary covering of side 2 and the classical $\Pi(2) \le 7$, this would yield $S(6) = 2$ and therefore $\Pi(2) = 7$ by (2). The public argument remains a candidate pending independent verification. □

12. Computational certificates

The geometric narrative of Sections 3–11 is supported by three layers of evidence.
Each certificate supports a specified implication chain:

$$\text{actual cover} \implies \text{actual points and geometric premises}$$
$$\implies \text{closed parameter domain} \implies \text{certified sign} \implies \bot.$$

Arithmetic checks the sign on its encoded domain. It does not establish the earlier geometric arrows. A contradiction in a model is useful only after the actual cover is proved to imply that model. Before multiplying a rational inequality by a denominator, prove its sign on the actual domain. Before applying a two-line chord budget, establish that both intersections are nonempty.

Exact rational certificates. On a polynomial over $[0,1]^m$, the tensor Bernstein basis is nonnegative and sums to one. Nonnegative (resp. strictly positive) coefficients prove nonnegativity (resp. strict positivity). Parameter domains are mapped into rectangles by explicit rational substitutions after denominators are shown positive. Verifiers regenerate targets, rebuild basis-change identities and rational coefficients, rather than trusting a generator's claimed minimal coefficient. Module R8 additionally checks a complete prefix-tree cover; module R64 checks every polynomial on its fixed rectangles.

Continuous interval certificates. The R16 root box outwardly contains the necessary domain. Each internal node splits into children whose union recovers the parent; each leaf is excluded by exact fractional envelopes. Denominator positivity, monotonic substitution direction, and all leaves together with their boundaries are checked.

Real-algebraic UNSAT queries. Selected propositions are accompanied by QF_LRA or QF_NRA queries. Only UNSAT is accepted. Queries are classified as (E) exact certificates, (H) hand proofs with solver cross-check, or (S) dual-solver UNSAT (z3 and cvc5) not yet reduced to (E). Dual-wrapper spot checks on the M8–M11 inventory report pass on the expanded faithfulness sample shipped with the ancillary package. Finite floating-point sampling is never used as an exclusion certificate.

Formal verification boundary. The main geometric upper bound is not claimed to be an unconditional Lean theorem. The public code contains partial reduction and geometry interfaces. A separate real-coordinate supporting library has been compiled locally: actual role selection, midpoint classification, a side-two cover, and twenty-eight R64 polynomial positivity results. These supporting theorems do not discharge all geometric exclusions. No Boolean status field, conditional bridge, or supporting build is used as a premise of the geometric theorem.

Reproduction. The complete detailed English appendix is still in preparation; the cited source modules carry the detailed geometric arguments. The pinned public repository supplies the original checker code and exact certificate data. The source manifest records the version used here; the verification report records the selected checks actually replayed for this edition. These are arithmetic and integrity checks, not an independent proof of every geometric reduction. From a clean source checkout one may run:

```
cd artifacts/checkers && python3 check_round64.py && python3 check_round63.py
cd ../closure && python3 m3_branch_checker.py
# optional: cd ../../lean/S6Core && lake build
```

13. Related work

Soifer's JCTA note and Cover-Up Squared writings introduced $\Pi(n)$ [9, 8]. Januszewski's Monthly note established $\Pi(2) \geq 6$ [7]. Friedman–Paterson and Friedman's surveys record constructions and small-case discussions [6, 5]. Dósa–Lángi–Tuza proved several exact small values and boundary analogues and isolated the conjecture $S(6) = 2$ [4]; Theorem 1.1 is a candidate argument addressing that conjecture. Asymptotic packing-waste problems of Chung–Graham [2, 3] concern a different regime.

On computer-assisted proofs in geometry, we follow the AMS guidelines on independent checkability and durable availability of code and data [1].

14. Concluding remarks

If confirmed, Theorem 1.1 would determine the first previously open value of $S(n)$ beyond the classical small cases. The same case-analysis and certificate pipeline suggests approaches to neighbouring values (e.g. refinements of $\Pi(3)$ and boundary-covering variants $S_{\mathrm{bd}}(n)$ from [4]), though those questions lie outside the present paper.

Acknowledgements

Certificate archives and Lean scaffolding accompany the argument in the public repository. Production used computer assistance (including generative tools); the releaser does not claim mastery of every step. See `DISCLOSURE.md`.

Appendix A. The sole normalization protocol

Only stepwise cover-dominating replacement is used. There is no preliminary simultaneous bilateral clamp, and no claim that all later corner operations are identities. The central tile C and remaining non-corner tile F stay fixed in each local corner operation. The symbol W means a selected corner tile after the current host representative has been specified.

Proposition A.1 (Exact replacement rule)**.** *Replace one corner tile V_i in a covering by a unit tile V_i' with $V_i \cap T_\mu \subseteq V_i' \cap T_\mu$, keeping the other five tiles. The new family covers the same target. If V_i is neither C nor F, every predicate involving only the unchanged C, F is unchanged.*

Proof. A point formerly covered solely by V_i is still in V_i'. All other points retain an unchanged covering tile. The last statement is equality of the two unchanged sets, not an invariance statement for corner hosts. □

This rule alone does not preserve a true gap, an injective host choice, the near/far representative, or a corner's empty central chord. After an operation that can change a predicate, compute it on the new actual family. An elimination must handle every resulting case or give an explicit map to a previously established exclusion. Leaving the old branch is not, by itself, a contradiction.

For each use of a local operation the reviewer should find:

(1) the replaced tile and its exact old and new projection intervals;
(2) target-relative containment and the actual points retained;
(3) all possible new branches with the exact earlier exclusions used;
(4) the surviving branch's strict inequalities and denominator signs.

The stable R identifiers locate these obligations and their arithmetic evidence. They do not replace a premise-transfer proof. The optional one-shot scheme is absent from this edition and cannot be used as an input in the active proof route. Its old status statements remain archival, outside the English reading copy.

Appendix B. Closure patch M2: frozen module R29N

The corrected round-29 continuation is frozen as module **R29N**. Earlier bounds with swapped parallel directions are withdrawn. The corrected R29N section 5 floating-chord bound is retained; it is not the withdrawn argument.

Proposition B.1 (R29N.1: three outer-edge states)**.** *Under the far-end hypothesis of Proposition 7.1 with* TOP *abutting, the remaining outer-edge states are exactly: (i) LEFT gap owned by C, RIGHT and TOP abutting; (ii) RIGHT gap owned by C, LEFT and TOP abutting; (iii) LEFT, RIGHT and TOP all abutting.*

Proposition B.2 (R29N.2: nonempty $F \cap H$ forces C off BOTTOM)**.** *If $F \cap H_0 \neq \varnothing$, then C avoids* BOTTOM *entirely.*

Proposition B.3 (R29N.4: triple abutment + nonempty $F \cap H$)**.** *Under triple abutment and $F \cap H_0 \neq \varnothing$, the free corner W cannot contain the top mid. (Does not migrate to lateral-gap cases.)*

Full inequality chains, including the 266-coefficient adjacent-corner certificates retained from R13, are in `REVIEW_OBLIGATIONS.md` and the long proof.

Appendix C. Closure patch M3: explicit branch tables

Class	Definition	Closing modules	Exhaustion note
G5	$F \ni O$; $k \in \{0,1,2\}$	R2–R8	Prop. 5.1
G1	unique centre, $\kappa = 0$	R9–R10C	Prop. 5.3
centre-G2	$\kappa = 1$, mid in C	R11–R12	Prop. 5.4
G4	$\kappa = 2$, adjacent mids	R20–R23	Prop. 5.5
G3	$\kappa = 2$, opposite mids	R24–R25	Prop. 5.6
residual G2	$\kappa = 1$, mid only in F	R13–R39; R44–R64	Cor. 5.7

Table 1. Table M3.0 (top-level). Full near-end / far-end / joining tables: `REVIEW_OBLIGATIONS.md`.

The Python checker `artifacts/closure/m3_branch_checker.py` verifies that the near-end Boolean grid (24 cells) is covered by live closing rows plus explicitly impossible cells. It is *not* a geometric proof. Running it on the shipped tree returns `PASS`.

Appendix D. Closure patch M4: E/H/S certificate tiers

Computational exclusions in the long proof are tagged:

(E) exact certificates (Bernstein / Farkas / rational IBB / CAD);
(H) hand proofs with solver cross-check;
(S) dual-solver UNSAT (z3 and cvc5) without an independent exact certificate.

The pinned public inventory records 151 checkable exact or hand-proof certificates and zero remaining (S) entries in that tracked upgrade inventory. This is not a statement that every geometric premise or every possible solver query has been independently verified. Inspect the geometric model before using any arithmetic conclusion. The supplied verification report states which checkers were actually replayed for this English edition.

References

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2. Fan Chung and Ron Graham, *Packing equal squares into a large square*, Journal of Combinatorial Theory, Series A **116** (2009), no. 6, 1167–1175, Asymptotic packing waste $W(x)$; cash challenges do not apply to $S(6) = 2$.
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4. György Dósa, Zsolt Lángi, and Zsolt Tuza, *Covering a square by congruent squares*, arXiv:2601.16535. https://arxiv.org/abs/2601.16535, 2026, States Conjecture 1.1 that $S(6) = 2$ (equivalently $\Pi(2) = 7$) without proof; proves several exact small values and boundary-covering analogues.
5. Erich Friedman and David Paterson, *Covering squares with unit squares*, https://erich-friedman.github.io/papers/covering/covering.html, Author-hosted version; accessed 2026-10-21.
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7. Janusz Januszewski, *A note on covering a square of side length* $2+\varepsilon$ *with unit squares*, The American Mathematical Monthly **116** (2009), no. 2, 174–178.
8. Alexander Soifer, *Cover-up squared*, Geombinatorics (2005), Series of notes on covering a square of side $n+\varepsilon$ with unit squares; see also Open Problem Garden entry "Covering a square with unit squares".
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